

Abhiman Singh Saharan

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EDUCATION

Bennett University

Bachelor of Technology in Computer Science Engineering

CGPA: **8.35/10**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

Greater Noida, India

2023 – 2027

BITS School of Management (BITSOM)

Product Management with Generative & Agentic AI

Mumbai, India

2026

PROJECTS

• InterviewIQ (AI-Powered Coding Interview Platform)

GitHub

- Built a full-stack AI-powered coding interview platform using React, TypeScript, Node.js, Express.js, PostgreSQL, and Prisma.
- Designed RESTful APIs for authentication, interview management, and adaptive interview generation using the Gemini API.
- Integrated Monaco Editor to enable real-time code execution, embedding speech analytics via Web Speech API to evaluate user communication metrics with less than 900ms latency.
- Developed a RAG-based system that personalizes interview questions using uploaded resumes and job descriptions.

TradeFlow (Institutional-Grade Electronic Trading Platform)

GitHub

- Built a full-stack electronic trading platform featuring a low-latency **price-time priority matching engine** supporting advanced order types, real-time order books, and institutional trading workflows.
- Designed an **event-driven backend architecture** using Redis Pub/Sub, Kafka, and WebSockets to decouple order matching from settlement, audit logging, and live market data streaming.
- Developed a native **FIX 4.4 protocol gateway**, pre-trade risk engine, historical market replay, and a GBM-based market simulator using Worker Threads for realistic trading simulations.
- Engineered a modern trading dashboard with portfolio management, risk analytics, watchlists, system telemetry, and real-time market visualization.

• AI Traffic Light Control System

GitHub

- Developed an adaptive traffic signal control system using OpenCV and Reinforcement Learning to optimize traffic flow.
- Implemented real-time vehicle detection and classification for cars, bikes, buses, and trucks using computer vision.
- Applied Q-Learning to dynamically optimize traffic signal timings based on real-time traffic density by 30%.
- Processed live video streams using OpenCV to extract traffic density metrics for adaptive signal control.

EXPERIENCE

CodSoft

Web Developer Intern

Remote

Jan 2026 – Feb 2026

- Developed responsive React.js applications, optimized frontend rendering by 22%, and ensured cross-browser compatibility.
- Built reusable React components and integrated RESTful APIs.
- Debugged production issues and improved application stability.
- Collaborated with a team of four developers using Agile methodologies.

SKILLS

- **Languages:** C++, Python, JavaScript, TypeScript, SQL
- **Frameworks & Technologies:** React.js, Next.js, Node.js, Express.js, FastAPI, PostgreSQL, SQLite, Prisma ORM
- **Tools:** Git, GitHub, Linux, Postman
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, System Design
- **AI/ML:** Gemini API, OpenAI API, OpenCV, TensorFlow, RAG

CERTIFICATIONS

Certified Ethical Hacker (CEH)

EC-Council(2026)

Fundamentals of Accelerated Computing with CUDA

NVIDIA Deep Learning Institute